

TESTs - Lesson 15 - NodeJS API using NestJS framework

NestJS Framework

- Building services with node
- Web server with node
- Using Express to run a node web
- About using frameworks
- Using NestJS framework
- Overview of a NestJS project
- Using the CLI
- Initializing a project with NestJS
- Swagger plugin

References

What is the primary purpose of an API?

- A To provide a graphical user interface for human interaction.
- B To enable machines, bots, or servers to communicate programmatically with an application.**
- C To store sensitive user data directly on the blockchain.
- D To manage frontend styling and layout for web pages.

When viewing an NFT marketplace like OpenSea, where does the displayed NFT information (prices, images, etc.) primarily come from?

- A Directly fetched from the blockchain by the frontend.
- B Retrieved from a centralized backend via API calls.**
- C Stored in the user's browser cache.
- D Generated dynamically by AI on the client side.

Why might you use a framework like NestJS instead of coding a Node.js server from scratch?

- A Frameworks are only for frontend development.
- B Coding a server from scratch is too simple and inefficient.
- C Frameworks handle boilerplate code for routing, security, and authentication, simplifying development.**
- D NestJS is the only framework compatible with TypeScript.

After running `npm run start` for the newly created NestJS project, accessing `localhost:3000` initially returns "Hello World!". What does this response signify?

- A The server is serving an HTML page.
- B The server is ready to accept user interface interactions.
- C The API is running and returning a plain text response for machine consumption.**
- D The project setup failed and returned a default error message.

In the analogy of "Manager / Worker" used for Controllers and Services, what is the primary role of the "Worker" (Service)?

- A To manage incoming work requests and delegate tasks.
- B To handle the overall business logic and orchestrate complex operations.
- C To receive a specific task from the manager and perform the detailed work.**
- D To make high-level decisions about the application's architecture.

What's the recommended way to load environment variables from a .env file in NestJS?

- A Directly hardcode the variables into the 'app.service.ts' file for easy access.
- B Use 'process.env' directly within any file without any specific import.
- C Install '@nestjs/config' and add 'ConfigModule.forRoot()' to your app module imports**
- D Add a "<script>" tag in the 'index.html' file to load the '.env' variables.

What is the main purpose of integrating Swagger (OpenAPI documentation) into a NestJS project?

- A To automatically generate all the necessary frontend components for the API.
- B To provide a visual, interactive documentation and testing interface for the API.**
- C To encrypt all API requests and responses for enhanced security.
- D To optimize API performance by reducing network latency.

What is the fundamental difference highlighted in the video between building UIs and APIs in Web3 applications?

- A UIs are built with JavaScript, while APIs are built with TypeScript.
- B UIs are for human interaction, while APIs facilitate machine-to-machine communication.**
- C UIs always store critical data on the blockchain, whereas APIs handle all data on the backend.
- D UIs are only used for frontend, and APIs are only for backend.

How does the NestJS framework simplify backend development, particularly regarding architectural organization and code reusability?

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PLAYLIST (10)



- A It eliminates the need for any services or controllers, directly connecting the frontend to the database.
- B It uses decorators and a modular approach, allowing developers to focus on business logic while abstracting away boilerplate code.**
- C It forces all backend code to be written in a single monolithic file, ensuring simplicity.
- D It exclusively supports JavaScript, which makes it easier to write complex backend logic.

Why should you handle sensitive data operations on the backend in Web3 apps rather than through frontend UI?

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PLAYLIST (10)



- A Frontend UIs are always slower for these operations.
- B Backend APIs provide better visual feedback to the user during sensitive transactions.
- C Frontend UIs inherently expose private keys and other sensitive data to the end-user, creating security vulnerabilities.**
- D Backend APIs automatically validate all blockchain transactions without needing extra code.